# Lecture-11

# Acid Base

Book Reference

**'General Chemistry'**

By Ebbing & Gammon

Ninth edition

Essentials of Physical Chemistry

Ball Tuli

# Acid-Base Reactions

- Acids and bases are some of the most important electrolytes.
- **Acids** have a sour taste, change litmus from blue to red, pH value 0 to below 7 and evolve $H_2$ gas upon reaction with active metals (such as alkali metals, alkaline earth metals, Zn, Al).
- Solutions of **bases**, on the other hand, have a bitter taste, change red litmus back to blue, pH value above 7 to 14, and a soapy feel.

# Continue..

- Acid react with bases to form salt and water.
- Their aqueous (water) solutions conduct electric current.
- Some examples of acids are acetic acid, present in vinegar; citric acid, a constituent of lemon juice; and hydrochloric acid, found in the digestive fluid of the stomach.
- An example of a base is aqueous ammonia, often used as a household cleaner (Table 2).

# TABLE 1. Common Acids and Bases

| *Name* | *Formula* | *Remarks* |
|---|---|---|
| **Acids** | | |
| Acetic acid | $HC_2H_3O_2$ | Found in vinegar |
| Acetylsalicylic acid | $HC_9H_7O_4$ | Aspirin |
| Ascorbic acid | $H_2C_6H_6O_6$ | Vitamin C |
| Citric acid | $H_3C_6H_5O_7$ | Found in lemon juice |
| Hydrochloric acid | $HCl$ | Found in gastric juice (digestive fluid in stomach) |
| Sulfuric acid | $H_2SO_4$ | Battery acid |
| **Bases** | | |
| Ammonia | $NH_3$ | Aqueous solution used as a household cleaner |
| Calcium hydroxide | $Ca(OH)_2$ | Slaked lime (used in mortar for building construction) |
| Magnesium hydroxide | $Mg(OH)_2$ | Milk of magnesia (antacid and laxative) |
| Sodium hydroxide | $NaOH$ | Drain cleaners, oven cleaners |

# Definition of Acids & Bases:

- Three concepts acid and base are available.
- 1. Arrhenius concept
- 2. Bronsted-Lowry concept
- 3. Lewis concept

# Arrhenius concept (1884)

- It is based on ionic dissociation of compound in water.
- An **acid** is a compound that releases $H^+$ ions in $H_2O$**.** For example, HCl in an acid since it releases $H^+$ ion when dissolved in $H_2O$.
- A **base** is a compound that releases $OH^-$ ions in $H_2O$. For example, NaOH is a base since it releases $OH^-$ ion when dissolved in $H_2O$.
- $HCl + H_2O \rightarrow H_3O^+ + Cl^-$
- $NaOH + H_2O \rightarrow Na^+ + OH^-$

# Continue..

- ***Usefulness***: This concept is useful in the study of chemical reactions.
- ***Limitations***:
- (a) Free $H^+$ ions do not exist in water.
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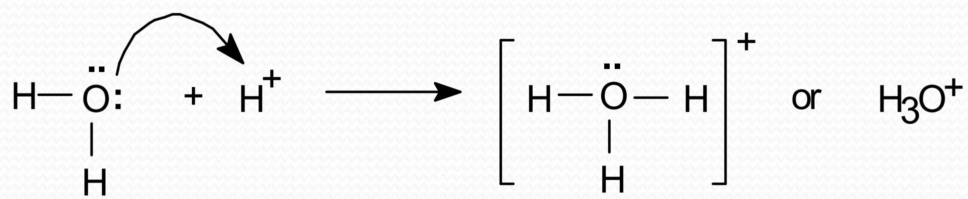


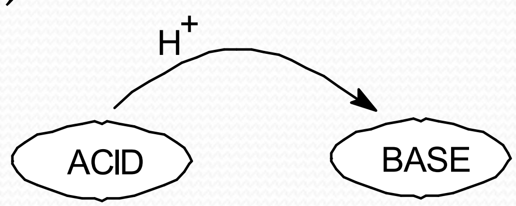

- Hydronium ion

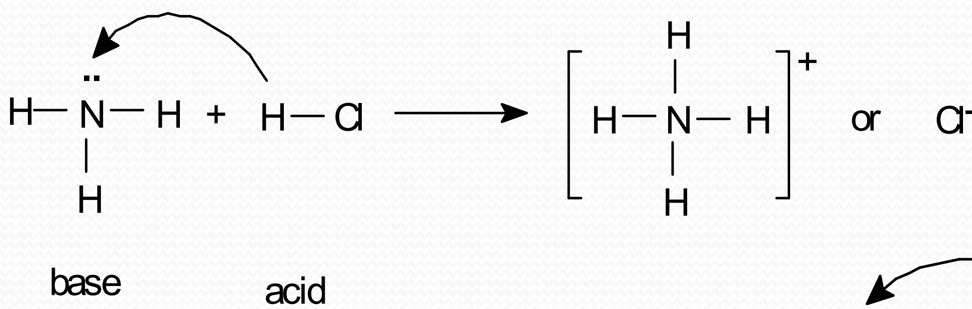
(b) Limited to water only: These definitions are applicable to water only.

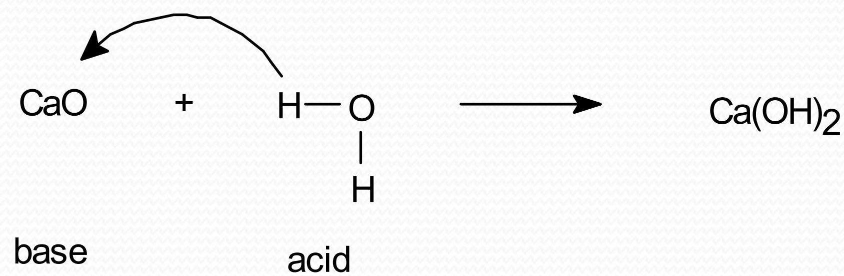
- (c) Some bases do not contain $OH^-$. Example: $NH_3$, CaO

## **Bronsted-Lowry concept** (1923)

- An acid is any molecule or ion that can donate a proton ($H^+$). A base is any molecule or ion that can accept a proton ($H^+$).



*Bronsted-Lowry concept is superior to Arrhenius concept*:

- (a) Much wider scope.
- (b) Not limited to aqueous solutions.

$$NH_3 + HCl \rightleftharpoons NH_4^+ + Cl^-$$

- (c) Release of $OH^-$ not necessary to qualify as a base.

$$NH_3 + H^+ \rightleftharpoons NH_4^+$$

base acid

"An acid is a proton donor, while a base is a proton acceptor."

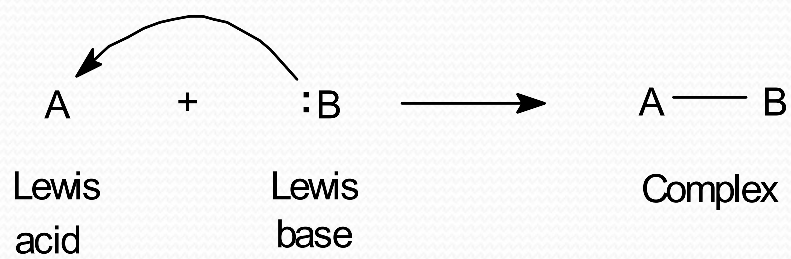
## Amphiprotic substances:

- Molecule or ions that behave both as Bronsted acid and Bronsted base is said to be amphiprotic. For example, $H_2O$.
- $HCl + H_2O \longrightarrow H_3O^+ + Cl^-$
- acid base
- $NH_3 + H_2O \longrightarrow NH_4^+ + OH^-$

  base acid
- Other Examples: $HS^-$, $Al(H_2O)_3(OH)_3$ etc. (both with $OH^-$ as acid, with $H_3O^+$ as base)

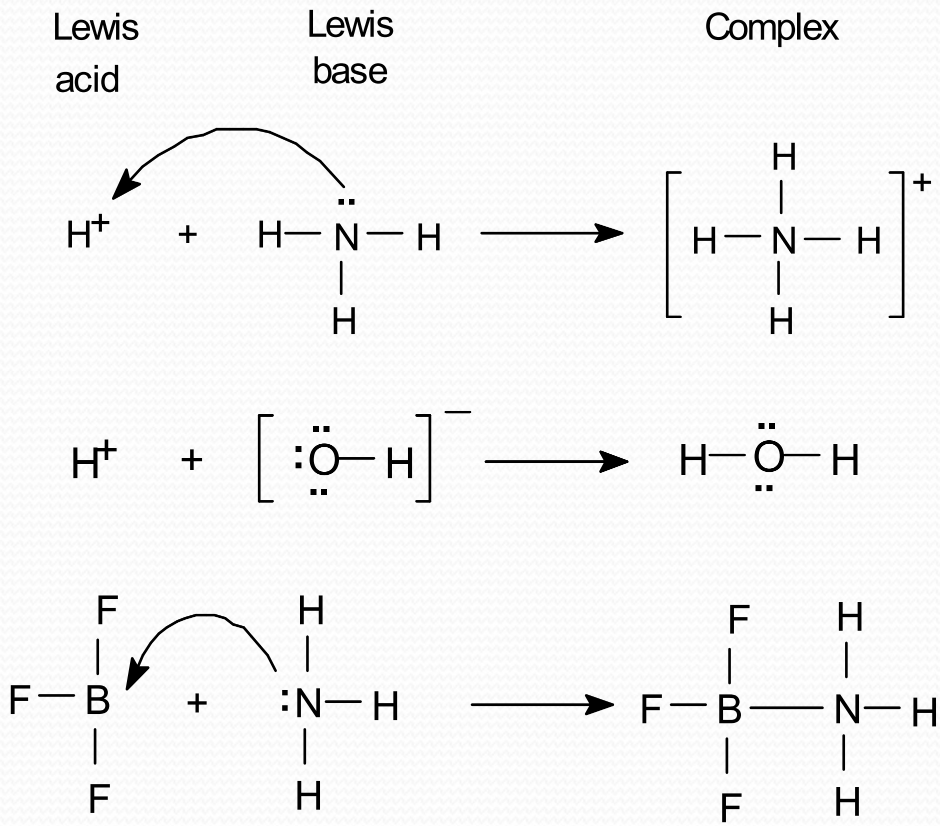
# Lewis concept (1930)

- An acid is a substance (molecule or ions) which can accept a pair of electrons. A base is a substance (molecule or ions) which can donate a pair of electrons.




- The combination of Lewis acid and Lewis base is called a complex. All cations or molecules short of an electron pair act as Lewis acids; and all anions or molecules having a lone pair of electron act as Lewis bases.

# Examples:



# Strong and Weak Acids and Bases

- Acids and bases are classified as strong or weak, depending on whether they are strong or weak electrolytes. A **strong acid** is *an acid that ionizes completely in water; it is a strong electrolyte.* Hydrochloric acid, HCl (*aq*), and nitric acid, $HNO_3$ (*aq*), are examples of strong acids.
-
- A **weak acid** is *an acid that only partly ionizes in water; it is a weak electrolyte.* Examples of weak acids are hydrocyanic acid, HCN (*aq*), and hydrofluoric acid, HF (*aq*).

# Continue..

- A **strong base** is *a base that is present in aqueous solution entirely as ions, one of which is $OH^-$; it is a strong electrolyte.* The ionic compound sodium hydroxide, NaOH, is an example of a strong base. It dissolves in water as $Na^+$ and $OH^-$. The hydroxides of Groups IA and IIA elements, except for beryllium hydroxide, are strong bases (see Table 3).
-
- A **weak base** is *a base that is only partly ionized in water; it is a weak electrolyte.* Ammonia, $NH_3$, is an example.

## TABLE 3. Common Strong and Weak Acids and Bases

| **Strong Acids** | **Strong Bases** | **Weak Acids** | **Weak Bases** |
|---|---|---|---|
| $HClO_4$ | $LiOH$ | $HCN$ | $NH_3$ |
| $H_2SO_4$ | $NaOH$ | $HF$ | $Mg(OH)_2$ |
| $HI$ | $KOH$ | $CH_3COOH$ | $Al(OH)_3$ |
| $HBr$ | $Ca(OH)_2$ | $H_2CO_3$ | $Na_2CO_3$ |
| $HCl$ | $Sr(OH)_2$ | $H_3PO_4$ | $NaHCO_3$ |
| $HNO_3$ | $Ba(OH)_2$ | $HCOOH$ | $CH_2NH_2$ |
| | | $HNO_2$ | $C_5H_5N$ |

- Non-metal oxides are acidic. Example: $SO_3 + H_2O = H_2SO_4$
- Metal oxides are basic. Example: $CaO + H_2O = Ca(OH)_2$

# Neutralization Reactions

- One of the chemical properties of acids and bases is that they neutralize one another. A **neutralization reaction** is *a reaction of an acid and a base that results in an ionic compound and possibly water.* When a base is added to an acid solution, the acid is said to be neutralized. *The ionic compound that is a product of a neutralization reaction* is called a **salt.** Most ionic compounds other than hydroxides and oxides are salts.
- $2HCl\ (aq) + Ca(OH)_2\ (aq) \rightarrow CaCl_2\ (aq) + H_2O\ (l)$
- Acid Base Salt
-
- $HCN\ (aq) + KOH\ (aq) \rightarrow KCN\ (aq) + H_2O\ (l)$
- Acid Base Salt

## Ionic Product of Water:
## Ionization of Water

Water is known to be slightly ionized,

$$H_2O \leftrightarrows H^+ + OH^-$$

But $H^+$ ions get hydrated to $H_3O^+$ ions by water, acting as a base. Water act as an acid by losing $H^+$ ion. Thus

$$H_2O + H_2O \leftrightarrows H_3O^+ + OH^-$$

# Ionization of Water (contd.)

Applying law of chemical equilibrium,

$$K = \frac{[H^+][OH^-]}{[H_2O]}, \quad \text{or} \quad K = \frac{[H_3O^+][OH^-]}{[H_2O][H_2O]}$$

In case of dilute solution,

$K\,[H_2O] = [H^+]\,[OH^-]$, or $K\,[H_2O]^2 = [H_3O^+]\,[OH^-]$

Since the ionic concentrations are very small, the concentration of unionized water may be taken as constant, thus

$$K\,[H_2O] = K_w, \qquad \text{or} \qquad K\,[H_2O]^2 = K_w$$

# Ionization of Water (contd.)

$$\therefore \; K_w = [H^+][OH^-] = [H_3O^+][OH^-]$$

$K_w$ is known as ionic product of water and may be defined as the product of concentration of $H^+$ ions and $OH^-$ ions in pure water. It is constant at constant temperature.

At $25^0C$, the value of $K_w$ is $1 \times 10^{-14}$.

# Ionization of Water (contd.)

In case of pure water and also in the case of neutral solutions, the molar concentration of $H^+$ ions and $OH^-$ ions are equal.

$$[H^+] = [OH^-] = \sqrt{(1 \times 10^{-14})} = 1 \times 10^{-7} \text{ moles/litre}$$

For neutral solution $[H^+] = [OH^-] = \sqrt{(1 \times 10^{-14})} = 1 \times 10^{-7}$ moles/litre.

For acidic solution $[H^+] > 1 \times 10^{-7} > [OH^-]$ moles/litre.

For basic solution $[OH^-] > 1 \times 10^{-7} > [H^+]$ moles/litre.

## pH value:

The acidity or the basicity of a solution can be expressed in terms of hydrogen ion concentration.

$$[H^+] = \frac{K_w}{[OH^-]} = \frac{1\times10^{-14}}{[OH^-]} \quad \text{and } [OH^-] = \frac{K_w}{[H^+]} = \frac{1\times10^{-14}}{[H^+]}$$

pH of a solution is the negative logarithm of hydrogen ion concentration (called pH scale, Sorensen in 1909).

# pH value (contd.)

$$p^{H} = -\log_{10}[H^{+}] = \log_{10}\frac{1}{[H^{+}]} \qquad Or, [H^{+}] = 10^{-pH}$$

$$p^{OH} = -\log_{10}[OH^{-}] = \log_{10}\frac{1}{[OH^{-}]}$$

For pure water, $[H^{+}] = [OH^{-}] = 1 \times 10^{-7}$ moles/litre

$$\therefore \quad pH = -\log_{10}[1\times 10^{-7}] = -(-7) = 7$$

$$\therefore \quad pOH = -\log_{10}[1\times 10^{-7}] = -(-7) = 7$$

$$\boxed{\therefore \quad pH + pOH = 7 + 7 = 14}$$

## Example-2:

# Calculate the pH of 0.001 M HCl.

## Solution:

HCl is a strong acid and it is completely dissociated in aqueous solution.

| | | | | |
|---|---|---|---|---|
| HCl | $\leftrightarrows$ | $H^+$ | + | $Cl^-$ |
| 0.001 M | | 0.001 M | | |

For every molecule of HCl, there is one $H^+$, therefore

$$[H^+] = [HCl]$$

# Solution (contd)

or $[H^+] = 0.001\ M$

$\therefore$ $pH = -\log(0.001)$

$= -\log(1 \times 10^{-3})$

$= -\log 1 + 3 \log 10$

$= 3$

Therefore the pH of 0.001 M HCl is 3. **Ans.**

## **Problem 3**. Calculate pH and pOH of 0.02 M $H_2SO_4$ solution. $K_w = 1 \times 10^{-14}$ at $25^0C$.

- $H_2SO_4 \leftrightarrow 2H^+ + SO_4^-$
- *equilib. conc.* 1M 2M 1M

$(2H_3O^+)$

If 1M $H_2SO_4$ solution ionizes completely, $[H_3O^+]$ will be 2M.

- Therefore, in a 0.02 M $H_2SO_4$ solution

$[H_3O^+] = 0.04$ M

- $\therefore [OH^-] = K_w / [H_3O^+] = (1 \times 10^{-14}) / 0.04 = 2.5 \times 10^{-13}$ M
- $\therefore pH = -\log [H_3O^+] = -\log (0.04) = 1.40$
- $\therefore pOH = -\log [OH^-] = -\log (2.5 \times 10^{-13}) = 12.60$ Ans.

**<u>Problem -4</u>. pH of an aqueous solution of HCl is 2.699 at 25$^0$C. Calculate the molarity of the solution.**

- We know from the definition of pH,
- $$\mathrm{pH} = -\log_{10}[H^+] = \log_{10}\frac{1}{[H^+]}$$
- $\therefore$ 2.699 = -log [$H_3O^+$]
- or, [$H_3O^+$] = antilog (- 2.699) = 0.002 M
- As HCl is a strong acid, it will ionize completely in the aqueous solution. So the molarity of HCl in the solution will be equal to the concentration of $H_3O^+$.
- $\therefore$ Molarity of HCl in the solution is 0.002. <u>Ans</u>.

**The pH of a solution of HCl is 2. Find out the amount of acid present in a litre of the solution.**

**pH = 2**

$-\log_{10}[H^+] = 2$

**The dissociation of HCl takes place according to the equation:**

$$HCl \longrightarrow H^+ + Cl^-$$

**One molecule of HCl gives one ion of $H^+$, Therefore,**

$[H^+] = [HCl] = 10^{-2}M$

**Amount of HCl in one litre = $10^{-2}$ x mol mass of HCl**

**= $10^{-2}$ x 36.5 = 0.365 $gl^{-1}$**

**SOLVED PROBLEM 1.** Determine the pH of 0.10 M NaOH solution.

**SOLUTION**

NaOH is a strong base and it is completely dissociated in aqueous solution.

$$\underset{0.10M}{NaOH} \rightleftharpoons Na^{+} + \underset{0.10M}{OH^{-}}$$

Therefore the concentration of $OH^-$ ions is equal to that of the undissociated NaOH.

$$[OH^-] = [NaOH] = 0.10\,M$$

$[H^+]$ can be calculated by applying the expression by substituting the value of $K_w = 1 \times 10^{-14}$ and of $[OH^-]$

$$[H^+] = \frac{K_w}{[OH^-]} = \frac{1 \times 10^{-14}}{0.10} = \frac{1 \times 10^{-14}}{10^{-1}} = 10^{-13}\,M$$

By definition

$$pH = -\log [H^+] = -\log^{-13}$$
$$= 13$$

Therefore the pH of 0.10 M NaOH is **13**.

**SOLVED PROBLEM.** Calculate the pH of 0.1 M $NH_3$ solution. The ionisation constant, $K_b$, for $NH_3$ is $1.8 \times 10^{-5}$.

**SOLUTION**

The equilibrium reaction is

$$\underset{0.10 - x}{NH_3} + H_2O \rightleftharpoons \underset{x}{NH_4^+} + \underset{x}{OH^-}$$

We assume $[OH^-] = x$ when equilibrium is reached. Thus we can write the equilibrium expression as

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]} = \frac{x^2}{0.10 - x}$$

Since $x$ is insignificant relative to 0.1, we can write

$$K_b = \frac{x^2}{0.10}$$

Solving for $x$,

$$x = \sqrt{0.10 \times K_b} \quad where \quad K_b = 1.8 \times 10^{-5}$$

$$x = \sqrt{0.10 \times (1.8 \times 10^{-5})} = 1.34 \times 10^{-3}\text{ M}$$

Since $x = [OH^-]$, we have

$$pOH = -\log(1.34 \times 10^{-3}) = 2.87$$

We know that

$$pH + pOH = 14.00$$

$$\therefore \quad pH = 14.00 - 2.87 = 11.13$$

Thus the pH of 0.1 M $NH_3$ solution is **11.1**.

# Handerson Equation

## CALCULATION OF THE pH OF BUFFER SOLUTIONS

The pH of an acid buffer can be calculated from the dissociation constant, $K_a$, of the weak acid and the concentrations of the acid and the salt used.

The dissociation expression of the weak acid, HA, may be represented as

$$HA \rightleftharpoons H^+ + A^-$$

and

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

or

$$[H^+] = K_a \times \frac{[HA]}{[A^-]} \quad ...(1)$$

The weak acid is only slightly dissociated and its dissociation is further depressed by the addition of the salt ($Na^+A^-$) which provides $A^-$ ions (Common ion effect). As a result the equilibrium concentration of the unionised acid is nearly equal to the initial concentration of the acid. The equilibrium concentration $[A^-]$ is presumed to be equal to the initial concentration of the salt added since it is completely dissociated. Thus we can write the equation (1) as

$$[H^+] = K_a \times \frac{[\text{acid}]}{[\text{salt}]} \qquad ...(2)$$

where [acid] is the initial concentration of the added acid and [salt] that of the salt used.

Taking negative logs of both sides of the equation (2), we have

$$-\log[H^+] = -\log K_a - \log\frac{[\text{acid}]}{[\text{salt}]} \qquad ...(3)$$

But $\quad -\log[H^+] = \text{pH} \quad$ and $\quad \log K_a = \text{p}K_a$

Thus from (3) we have

$$\text{pH} = \text{p}K_a - \log\frac{[\text{acid}]}{[\text{salt}]} = \text{p}K_a + \log\frac{[\text{salt}]}{[\text{acid}]}$$

Hence $$\mathbf{pH = pK_a + \log\frac{[salt]}{[acid]}}$$

This relationship is called the **Henderson-Hasselbalch equation** or simply **Henderson equation.**

In a similar way, the Henderson-Hasselbalch equation for a basic buffer can be derived. This can be stated as :

$$\mathbf{pOH = pK_b + \log\frac{[salt]}{[base]}}$$

## Significance of the Henderson-Hasselbalch equation

With its help :

(1) The pH of a buffer solution can be calculated from the initial concentrations of the weak acid and the salt provided $K_a$ is given.

However, the Henderson-Hasselbalch equation for a basic buffer will give pOH and its pH can be calculated as (14 – pOH).

(2) The dissociation constant of a weak acid (or weak base) can be determined by measuring the pH of a buffer solution containing equimolar concentrations of the acid (or base) and the salt.

$$\mathrm{pH} = \mathrm{p}K_a + \log \frac{[\mathrm{salt}]}{[\mathrm{acid}]}$$

Since $[\mathrm{salt}] = [\mathrm{acid}], \quad \log \frac{[\mathrm{salt}]}{[\mathrm{acid}]} = \log 1 = 0$

$\therefore \quad \mathrm{p}K_a = \mathrm{pH}$

The measured pH, therefore, gives the value of p$K_a$ of the weak acid.

Likewise we can find the p$K_b$ of a weak base by determining the pH of equimolar basic buffer.

(3) A buffer solution of desired pH can be prepared by adjusting the concentrations of the salt and the acid added for the buffer.

It is noteworthy that buffer solution are most effective when the concentrations of the weak acid (or weak base) and the salt are about equal. This means that pH is close to the value of p$K_a$ of the acid (or p$K_b$ of the base).

# Buffer Solution

## WHAT IS A BUFFER SOLUTION ?

It is often necessary to maintain a certain pH of a solution in laboratory and industrial processes. This is achieved with the help of **buffer solutions, buffer systems or simply buffers.**

**A buffer solution is one which maintains its pH fairly constant even upon the addition of small amounts of acid or base.**

In other words, a buffer solution resists (or buffers) a change in its pH. That is, we can add a small amount of an acid or base to a buffer solution and the pH will change very little. Two common types of buffer solutions are :

(1) a weak acid together with a salt of the same acid with a strong base. These are called **Acid**

**buffers** *e.g.*, $CH_3COOH + CH_3COONa$.

(2) a weak base and its salt with a strong acid. These are called **Basic buffers.** *e.g.*, $NH_4OH + NH_4Cl$.

Let us illustrate buffer action by taking example of a common buffer system consisting of solution of acetic acid and sodium acetate ($CH_3COOH/CH_3COONa$).

$$CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$$

$$CH_3COONa \longrightarrow Na^+ + CH_3COO^-$$

since the salt is completely ionised, it provides the common ions $CH_3COO^-$ in excess. The common ion effect suppresses the ionisation of acetic acid. This reduces the concentration of $H^+$ ions which means that pH of the solution is raised. Thus, a 0.1 M acetic acid solution has a pH of 2.87 but a solution of 0.1 M acetic acid and 0.1 M sodium acetate has a pH of 4.74 (Fig. 27.11). Thus 4.74 is the pH of the buffer. On addition of 0.01 mole NaOH the pH changes from 4.74 to 4.83, while on the addition of 0.01 mole HCl the pH changes from 4.74 to 4.66. Obviously the buffer solution maintains fairly constant pH and the changes in pH could be described as marginal.

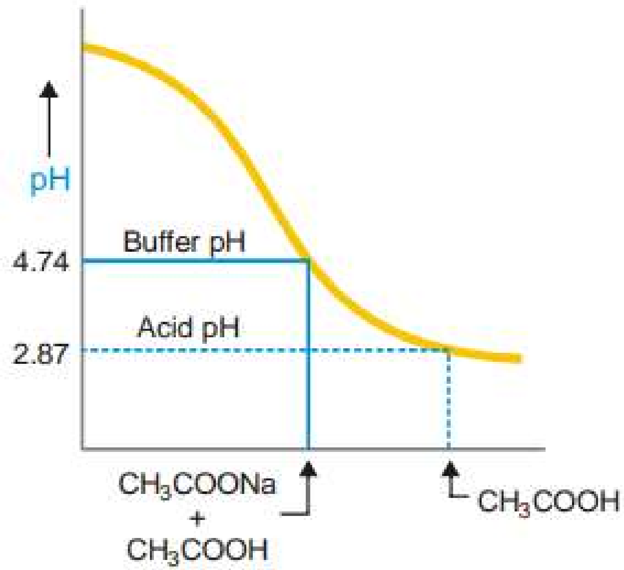



**Figure 27.11**
**The buffer solution ($CH_3COOH/CH_3COONa$) has a higher pH than the acid itself.**

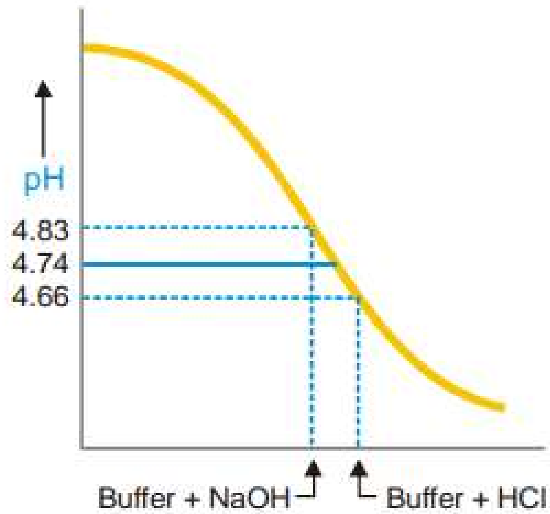
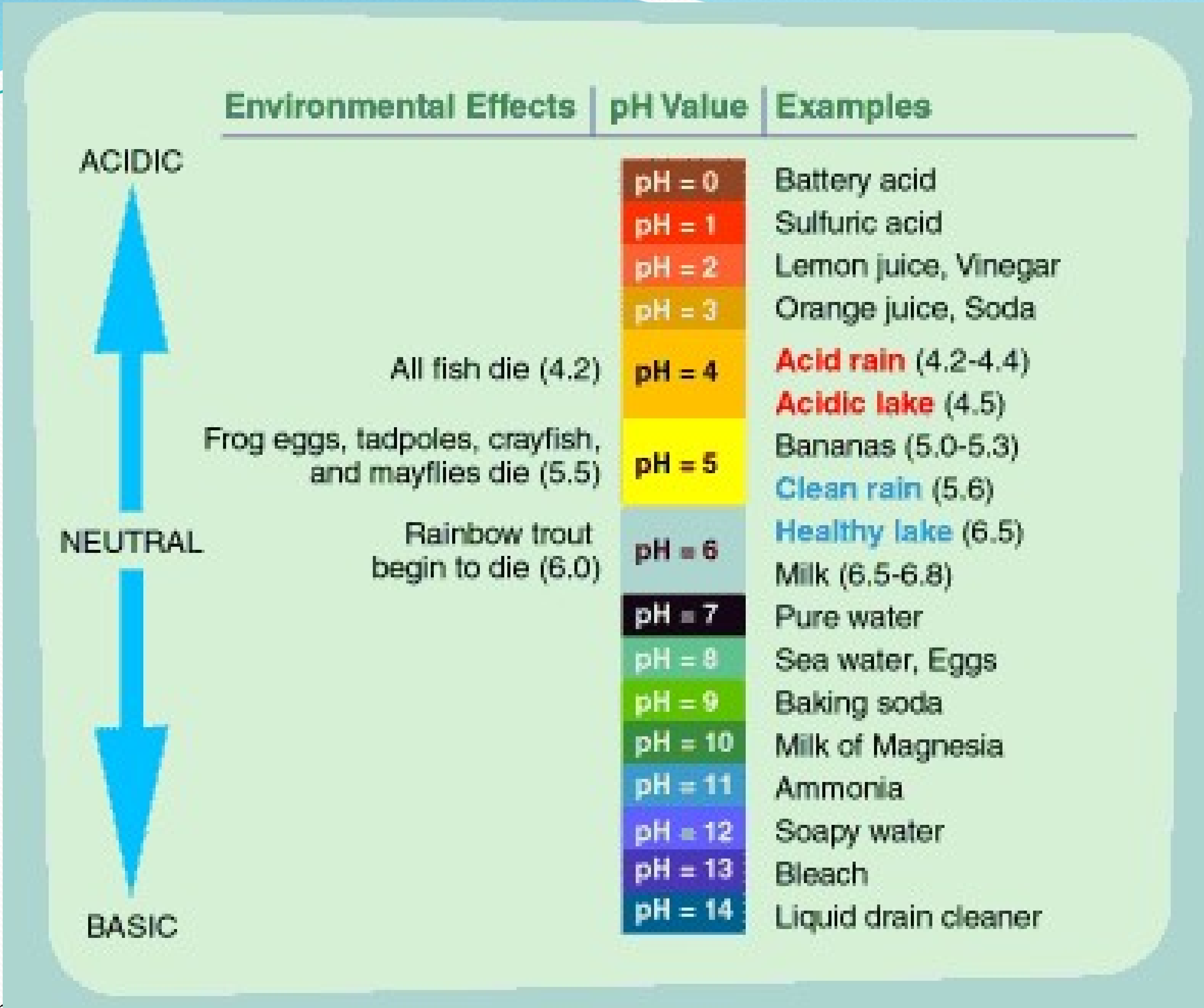



**Figure 27.12**
**The pH of buffer changes only slightly upon addition of an acid (0.01 moles) or base (0.01 mole NaOH).**

# Importance of pH in our daily Life

- **All living organisms are pH sensitive and can survive only in a narrow range of pH.**

- Most foods are slightly acidic; **our "bodily fluids" are slightly alkaline**, as is seawater— not surprising, since early animal life began in the oceans. The pH of freshly-distilled water will go downward as it takes up carbon dioxide from the air. "Acid" rain is by definition more acidic than pure water in equilibrium with atmospheric $CO_2$, owing mainly to sulfuric and nitric acids that originate from fossil-fuel emissions of nitrogen oxides and $SO_2$.
- pH of the soil: Plants require a specific pH range for their healthy growth.


Environmental Effects
pH Value
Examples
ACIDIC
NEUTRAL
BASIC
pH = 0
Battery acid
pH = 1
Sulfuric acid
pH = 2
Lemon juice, Vinegar
pH = 3
Orange juice, Soda
All fish die (4.2)
pH = 4
Acid rain (4.2-4.4)
Acidic lake (4.5)
Frog eggs, tadpoles, crayfish, and mayflies die (5.5)
pH = 5
Bananas (5.0-5.3)
Clean rain (5.6)
Rainbow trout begin to die (6.0)
pH = 6
Healthy lake (6.5)
Milk (6.5-6.8)
pH = 7
Pure water
pH = 8
Sea water, Eggs
pH = 9
Baking soda
pH = 10
Milk of Magnesia
pH = 11
Ammonia
pH = 12
Soapy water
pH = 13
Bleach
pH = 14
Liquid drain cleaner

## Importance of pH in our daily Life…

- **Plants and animals are pH sensitive**: Our body works within the pH range of 7.0 to 7.8. When pH of rain water is less than 5.6, it is called acid rain. When acid rain flows into the rivers, it lowers the pH of the river water. The survival of aquatic life in such rivers becomes difficult.

- **pH in our digestive system**: It is very interesting to note that our stomach produces hydrochloric acid. It helps in the digestion of food without harming the stomach. During indigestion the stomach produces too much acid and this causes pain and irritation. To get rid of this pain, people use bases called antacids. These antacids neutralise the excess acid.

## Importance of pH in our daily Life…

- **pH change as the cause of tooth decay**: Tooth decay starts when the pH of the mouth is lower than 5.5. Tooth enamel, made up of calcium phosphate is the hardest substance in the body. It does not dissolve in water, but is corroded when the pH in the mouth is below 5.5. Bacteria present in the mouth produce acids by degradation of sugar and food particles remaining in the mouth after eating. The best way to prevent this is to clean the mouth after eating food. Using toothpastes, which are generally basic, for cleaning the teeth can neutralize the excess acid and prevent tooth decay.

# Importance of pH in our daily Life…

- **Self defence by animals and plants through chemical warfare**: Wasps (insect) and jellyfish have an alkaline sting and bees have an acidic sting. So with wasps stung area can be treated with vinegar, and bees with soap or baking soda. Stinging hair of nettle (plant) leaves inject methanoic acid causing burning pain.

- **The balance of pH** in our body also helps to regulate our breathing rate (carbonic acid in our blood), controls microorganisms on skin, and activates enzymes. Blood has a pH which needs to be maintained between 7.35 and 7.45, or else serious illness and death may occur.

# Thank you All